



## AQ4.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

Choose the set of correct options.

☐

If  $S$  is a spanning set of the vector space  $V$ , then  $S \cup \{v\}$  must be a spanning set of  $V$ , for all  $v \in V$ .

☐

Span of an empty set is the zero vector space.

☐

If  $S$  is a spanning set of the vector space  $V$ , then  $S \setminus \{v\}$  must be a spanning set of  $V$ , for all  $v \in S$ .

☐

If  $S$  is a spanning set of the vector space  $V$ , then  $S \cup \{v\}$  may not be a spanning set of  $V$ , for all  $v \in V$ .

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

If  $S$  is a spanning set of the vector space  $V$ , then  $S \cup \{v\}$  must be a spanning set of  $V$ , for all  $v \in V$ .

Span of an empty set is the zero vector space.

Let  $S$  be the set of matrices where  $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ . Consider the following statements:

- Statement P:  $\text{Span}(S)$  is the vector space consisting of only lower triangular square matrices of order 2.
- Statement Q:  $\text{Span}(S)$  is the vector space consisting of only upper triangular square matrices of order 2.
- Statement R:  $\text{Span}(S)$  is the vector space consisting of all the square matrices of order 2.
- Statement S:  $\text{Span}(S)$  is the vector space consisting of only scalar matrices of order 2.

Find the number of correct statements.

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

(Type: Numeric) 1

1 point

If  $S$  is a basis of  $\mathbb{R}^2$ , then what is the cardinality of  $S$ ?

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

(Type: Numeric) 2

1 point



Let  $S = \{(1, 2), (1, 3)\}$   $S = \{(1, 2), (1, 3)\}$  and  $\text{span}(S) = R^n$   $\text{span}(S) = R^n$ . Then find the value of  $n$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 2

1 point

Let  $S$  be the set of matrices where  $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$   $S = \{[1000], [0001]\}$ . Consider the following

statements:

- Statements P:  $\text{Span}(S)$  is the vector space consisting of only the Identity matrix of order 2.
- Statements Q:  $\text{Span}(S)$  is the vector space consisting of only diagonal matrices of order 2.
- Statements R:  $\text{Span}(S)$  is the vector space consisting of all the square matrices of order 2.
- Statements S:  $\text{Span}(S)$  is the vector space consisting of only scalar matrices of order 2.

Find the number of correct statements.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

1 point

Consider the following set of vectors  $S = \{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$   $S = \{(1, 1, -1), (-1, 1, 1), (0, 2, 1, 0), (0, 1, -2), (1, 0, -2)\}$ . Choose the set of correct options.

☐

The set  $\{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0)\}$  is not a basis of  $R^3$ .

☐

The set  $\{(0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$  is a basis of  $R^3$ .

☐

The set  $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$  is a basis of  $R^3$ .

☐ None of the above.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The set  $\{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0)\}$  is not a basis of  $R^3$ .

The set  $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$  is a basis of  $R^3$ .

1 point

Consider a set of vectors  $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\}$   $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\}$  in  $R^2$ . Choose the set of correct options.

☐

The set  $\{(1, 2), (-1, 2), (3, 1)\}$  is a basis of  $R^2$ .

☐

The set  $\{(-1, 2), (3, 1)\}$  is a basis of  $R^2$ .

☐

The set  $\{(3, 1), (-3, 1)\}$  is a basis of  $R^2$ .

☐ None of the above.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The set  $\{(-1, 2), (3, 1)\}$  is a basis of  $R^2$ .

The set  $\{(3, 1), (-3, 1)\}$  is a basis of  $R^2$ .

## Level 2

1 point

Choose the set of correct options.

☐

If a subset  $SS$  of  $R^2$  contains only two elements then  $SS$  is a basis of  $R^2$ .

☐

If a subset  $SS$  of  $R^3$  contains only one elements then  $SS$  can never be a basis of  $R^3$ .

☐

If a subset  $SS$  of  $RR$  contains only one non zero element then  $SS$  is a basis of  $RR$ .

☐

If  $S_1$  and  $S_2$  are two bases of  $R^3$  then  $S_1 = S_2$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If a subset  $SS$  of  $R^3$  contains only one elements then  $SS$  can never be a basis of  $R^3$ .

If a subset  $SS$  of  $RR$  contains only one non zero element then  $SS$  is a basis of  $RR$ .

1 point

Let  $V$  be the subspace of  $R^3$  defined as follows:

$$V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in R\}$$

Choose the set of correct options from the following.

☐

$\{(1, 1, 0), (1, 0, -1)\}$  is a linearly independent set of  $V$ .

☐

$\{(1, 1, 0), (1, 0, -1), (0, 1, 1)\}$  is a linearly independent set of  $V$ .

☐

$\{(0, 1, 1), (1, 0, -1)\}$  is a spanning set of  $V$ .

☐

$\{(1, 1, 0)\}$  is a spanning set of  $V$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{(1, 1, 0), (1, 0, -1)\}$  is a linearly independent set of  $V$ .

$\{(0, 1, 1), (1, 0, -1)\}$  is a spanning set of  $V$ .

1 point

Let  $V$  and  $W$  be vector spaces which are defined as follows:

$$V = \{(x, y) \mid y = mx, \text{ where } m \neq 0 \text{ and } x, y, m \in R\}$$

$\{(x, y) \mid y = mx, \text{ where } m = 0 \text{ and } x, y, m \in R\}$  with usual addition and scalar multiplication as in

$R^2$ .  $W = \{(x, y) \mid x = 0\}$  with usual addition and scalar multiplication as in  $R^2$ .

$\{(x, y) \mid x = 0\}$  with usual addition and scalar multiplication as in  $R^2$ .

Choose the correct set of options.

☐

The set  $\{(1, m), (\frac{1}{m}, 1)\}$  is a linearly independent set in  $V$ .

☐

The set  $\{(1, m), (\frac{1}{m}, 1)\}$  is a spanning set for  $V$ .

☐

The set  $\{(1, m)\}$  is a linearly independent set in  $V$ .

☐

The set  $\{(\frac{1}{m}, 1)\}$  is a linearly independent set in  $V$ .

☐

The set  $\{(0, 1), (0, 2)\}$  is a linearly independent set in  $W$ .

☐

The set  $\{(0, 1)\}$  is a linearly independent set in  $W$ .

☐

The set  $\{(0, 5)\}$  is a spanning set for  $W$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The set  $\{(1, m), (\frac{1}{m}, 1)\}$  is a spanning set for  $V$ .

The set  $\{(1, m)\}$  is a linearly independent set in  $V$ .

The set  $\{(\frac{1}{m}, 1)\}$  is a linearly independent set in  $V$ .

The set  $\{(0, 1)\}$  is a linearly independent set in  $W$ .

The set  $\{(0, 5)\}$  is a spanning set for  $W$ .

[Check Answers](#)

Your score is: 0/10